

should not be used as a search variable; a user-initialized control variable that uses MFLOWJ should be used instead.

W4(I) Numeric part of variable request code. This is assumed 0 if missing. For example, to use the pressure in volume 567010000 for the search argument, the user would enter P for W3 and 567010000 in W4.

8.3.3 Time-Dependent Volume Data, CCC0201-0299

These cards are required for time-dependent volume components. A set of data is made up of the search variable (e.g., time) followed by the required data indicated by control Word 1 in card CCC0200. The card numbers need not be consecutive, but the value of the search variable in a succeeding set must be equal to or greater than the value in the previous set. One or more sets of data, up to 5,000 sets, are allowed. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. Linear interpolation is used if the search argument lies between the search variable entries. End-point values are used if the argument lies outside the table values. Only one set is needed if constant values are desired, and computer time is reduced when only one set is entered. Step changes can be accommodated by entering the two adjacent sets with the same search variable values or an extremely small difference between them. Given two identical argument values, the set selected will be the closest to the previous argument value. Sets may be entered one or more per card and may be split across cards. The total number of words must be a multiple of the set size.

Inputting time-dependent volume tables where the search variable is a thermodynamic variable from some other component can run into difficulties if the component numbering is such that the time-dependent volume is initialized before the component providing the needed search variable. A reliable fix for this is to make the search variable a control system output in the desired units, while the thermodynamic variable is the control system input in code internal (SI) units. The control system initial value can be set to the desired initial value of the search variable, and this will be used by the time-dependent table.

W1(R) Search variable (e.g., time).

W2-W7(R) Quantities as described under Word 1 in card 200. Depending on the control word, 2-5 quantities may be required. If entered, boron concentration (parts of boron per parts of liquid) follows the last required word for thermodynamic conditions.

As described above, sets may be entered one or more per card.

8.4 Single-Junction Component, SNGLJUN

A single-junction component is indicated by SNGLJUN for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the junction in the single-junction component is numbered CCC000000.

8.4.1 Single-Junction Geometry, CCC0101-0109

This card (or cards) is required for single-junction components.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. An old or an expanded format can be used to connect all other volumes. In the old format, use ccc000000 if the connection is to the inlet side of the component and use ccc010000 if the connection is to the outlet side of the component. In the expanded format, the connection code is cccvv000n, where ccc is the component number, vv is the volume number, and n indicates the face number. A nonzero n specifies the expanded format. The number n equal to 1 and 2 specifies the inlet and outlet faces, respectively, for the volume's coordinate direction. The number n equal to 3 through 6 specifies crossflow. The number n equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; n equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume using the old format, both CCC000000 and CCC010000 are allowed. For connecting to a time-dependent volume using the expanded format, only the number N equal to 1 or 2 is allowed.

When this junction is used to connect a "FROM" 3D volume with a "TO" 1D volume, the connection code is cccvv000n, where ccc is the channel number of the 3D volume, vv is the mesh number (.ge. 2), and n indicates the face number. The numbers n equal to 7 and 8 specify the bottom and top faces, respectively. The number n equal to 9 specifies the side face.

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

When this junction is used to connect a "FROM" 1D volume with a "TO" 3D volume, the connection code is cccvv000n. See the description for W1 above.

W3(R) Junction area (m^2 , ft^2). If zero, the area is set to the minimum volume flow area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or smaller than the minimum of the adjoining volume areas. For smooth area changes, there are no restrictions.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. Note: a variable loss coefficient may be specified. The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6. If the smooth area change option is selected ($a = 0$), the entry here is the loss

coefficient appropriate for the junction flow area resulting from the entry made in W3. If the abrupt area change option is selected ($a = 1$ or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction. If the $a = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

- W5(R) Reynolds number independent reverse flow energy loss coefficient A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. Note: a variable loss coefficient may be specified. See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros.
- j-flag The digit j specifies that this junction is a jet junction. Pool surface condensation is enhanced in the volume above the junction when this model is activated. This junction must be underneath the to volume.
- e-flag The digit e specifies the modified PV term in the energy equations. $\underline{e} = 0$ means that the modified PV term will not be applied, and $\underline{e} = 1$ means that it will be applied.
- f-flag The digit f specifies CCFL options. $\underline{f} = 0$ means that the CCFL model will not be applied, and $\underline{f} = 1$ means that it will be applied.
- v-flag The digit v specifies horizontal stratification entrainment/pullthrough options. This model is for junctions connected to a horizontal volume. $\underline{v} = 0$ means the model is not applied; $\underline{v} = 1$ means an upward-oriented junction (offtake volume must be vertical); $\underline{v} = 2$ means a downward-oriented junction (offtake volume must be vertical); and $\underline{v} = 3$ means a centrally (side) located junction; $\underline{v} = 4$ means extended angled side located junction.
- c-flag The digit c specifies choking options. $\underline{c} = 0$ means that the choking model will be applied, and $\underline{c} = 1$ means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag The digit a specifies area change options. $\underline{a} = 0$ means a smooth area change and only the user supplied K_{loss} is applied. $\underline{a} = 1$ means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. $\underline{a} = 2$ means a partial abrupt area change model and it is the same as

"a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two velocity momentum equations) option, and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1.

s-flag The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to volume and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 uses momentum flux in the to volume, but not in the from volume. s = 3 does not use momentum flux in either the to or the from volume.

W7(R) The meaning of this word is different, depending on which the critical flow model has been selected (see the c-flag under W6 above).

If the Henry-Fauske critical flow model is active, the discharge coefficient is entered. If W7 and W8 are missing, W7 is set to 1.0, and W8 is set to 0.14.

If the original RELAP5 critical flow model is active, the subcooled discharge coefficient is entered. This entry applies only to subcooled liquid choked flow calculations. The entry must be > 0.0 and < 2.0. If W7, W8, and W9 are missing, they are each set to 1.0.

W8(R) The meaning of this word is different, depending on which the critical flow model has been selected (see the c-flag under W6 above).

If the Henry-Fauske critical flow model is active, the thermal nonequilibrium constant is entered. If W7 is entered, and W8 is missing, W8 is set to 0.14. If W8 is < 0.01, the equilibrium option is used, and W8 is reset to 0.0. If W8 is >1000.0, the frozen option is used, and W8 is reset to 100.0.

If the original RELAP5 critical flow model is active, the two-phase discharge coefficient is entered. This entry applies only to two-phase choked flow calculations. The entry must be > 0.0 and < 2.0. If W7 is entered, and W8 and W9 are missing, W8 and W9 are each set to 1.0.

W9(R) The meaning of this word is different, depending on which the off-take model for an extended angled side junction has been selected (see the v-flag under W6 above) and which the critical flow model has been selected (see the c-flag under W6 above).

If -flag of W6(I) is 4, this word is the horizontal angle (degree) between from-volume and to-volume.

If the Henry-Fauske critical flow model is active, this word is not used and must not be entered. If it is entered, an input error results. This drastic action is to prevent the user from using an older deck with the new code without changing the three discharge coefficients from their values for the original RELAP5 critical flow model to an appropriate set for the Henry-Fauske critical flow model. W8 has a completely different meaning in the two models.

If the original RELAP5 critical flow model is active, the superheated discharge coefficient is entered. This entry applies only to superheated vapor choked flow calculations. The entry must be > 0.0 and < 2.0 . If W7 and W8 are entered, and W9 is missing, then W9 is set to 1.0.

8.4.2 Single-Junction Diameter and CCFL Data, CCC0110

This card is optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation (i.e., $\underline{f} = 0$ in Word 6 of cards CCC0101-0109), then, the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If this card is being used for the CCFL model (i.e., $\underline{f} = 1$ in Word 6 of cards CCC0101-0109), then enter all four words for the appropriate CCFL model if values different from the default values are desired.

- W1(R) Junction hydraulic diameter, D_j (m, ft). This quantity is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . This number should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0 is entered or if the default is used, the junction diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 3 of cards CCC0101-0109 for the area.
- W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).
- W3(R) Gas intercept, c . This quantity is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.
- W4(R) Slope, m . This quantity is the slope used in the CCFL correlation and must be > 0 . The default value is 1.

8.4.3 Single-Junction Form Loss Data, CCC0111

This card is optional. The user-specified form loss is given in Words 4 and 5 of card CCC0101 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of card CCC0101. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

8.4.4 Single-Junction Initial Conditions, CCC0201

This card is required for single-junction components.

W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flows.

W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) [$W1 = 0$] or mass flow (kg/s, lb/s) [$W1 = 1$].

W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) [$W1 = 0$] or mass flow (kg/s, lb/s) [$W1 = 1$].

W4(R) Interface velocity (m/s, ft/s). Enter 0.

8.5 Time-Dependent Junction Component, TMDPJUN

This component is indicated by TMDPJUN for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the junction in the time-dependent junction component is numbered as CCC000000.